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Enhanced Network Attack Detection Using a TCN–BiLSTM–Transformer Framework

Naguru Vijay Kumar
Dept. of IT

Narasaraopeta Engineering College
Narasaraopet, India
vijaycse.scet@gmail.com

Duggempudi Anji Reddy
Dept. of IT

Narasaraopeta Engineering College
Narasaraopet, India
anjireddyduggempudi@gmail.com

Dasari Sri Saran
Dept. of IT

Narasaraopeta Engineering College
Narasaraopet, India
dasarisan04@gmail.com

Mekala Karthik
Dept. of IT

Narasaraopeta Engineering College
Narasaraopet, India
Karthikmekalakarthik3@gmail.com

N. Rajasekhara
Dept. of Information Technology
GRIET

Hyderabad, Telangana, India
rajasekhara1581@grietcollege.com

Tippana Chalapathi Rao
Dept. of CSE

Aditya Institute of Technology and Management
Srikakulam, Andhra Pradesh, India.
chalapathit520@gmail.com

Abstract—The risk of network security is continuously growing in scope and complexity and therefore in need of intrusion detection systems, which would be capable of deriving meaningfulness out of types and dynamically changing traffic patterns. Previous methodologies have created shortcomings in several key areas which include, but are not limited to, subjected bias, spatiotemporal relationships, and the pattern recognition required for the analysis of large amounts of interconnected data. To tackle such issues, an exemplary work has presented in this case an architecture which includes the fusion of Convolutional Networks, Bidirectional Long Short Term models, and Transformer encoders. The integrated frameworks explain network traffic in the short-term, the long-term, and incorporate the global dependencies. The models are trained and tested in NSL-KDD and UNS-NB15 datasets, in the course of which various preprocessing techniques, i.e. one-hot encoding, feature permutation, sliding window-segmentation, and resampling procedures used to combat the category imbalance issue. Through an empirical investigation the advanced architectures have been compared to traditional machine learning and comparative deep learning and have been found to have outperformed these approaches in terms of accuracy, precision, and F-1 scores. The results indicate profound temporal detail extraction will considerable boost the efficiency and effectiveness of intrusion detection systems, and facilitate the detection of a wide variety of attack forms.

Index Terms—Network intrusion detection, temporality feature extraction, TCN, BiLSTM, Transformer encoder, network traffic analysis, class imbalance, feature selection, sliding-window, cybersecurity.

I. INTRODUCTION

A. Motivation

The fast growth of online networks has resulted in a rush of advanced cyberattacks that have the potential to circumvent traditional security controls. Recent network traffic is huge, unbalanced, and very dynamic and as such, traditional intrusion detection modules perform poorly, especially on complex and low frequency attacks. And while such low performance issues might be understood in terms of complex models, most models fail to capture even elementary short term and long term

temporal dependencies from the traffic sequence and as such, perform poorly on low detection accuracy. Given such challenges, the demand for a complex intrusion detection system that can learn rare detailed temporal behaviours, detect minor attack patterns, and can for all conditions remain responsive to a wide array of traffic patterns is critical. This project is based on the need to build a hybrid deep learning model from the frameworks of focusing on multiple time feature extractors to improve and capture a wide array of network intrusion detection to address the detection fine-tuning. This is because most of the current approaches aim to address the detection of long term stealthy intrusions. To the best of my knowledge, most of the existing algorithms are isolated, that is, they are frame synchronous and packet reuse unaware of long term dependencies between actions, making them fail to capture the full context of attack sequence.

B. Research Gap

The effectiveness of the current intrusion detection system has a number of weaknesses, which have been exacerbated by the development of intrusion detection system. Most of the existing models do not give long range temporal relationships in traffic since they focus more on isolated packet attributes instead of the relationship between events as they take place over time. This complicates the efforts to identify sneaky or multi-stage attacks. The other big gap is that it cannot deal with imbalanced datasets, with rare categories of attack often significantly undersampled, which makes traditional models ignore low-frequency and yet critically high-intensity threats. Moreover, the vast majority of studies are based only on one deep learning method, which means that they do not completely extract features as no single model is capable of capturing local features, long-distance interactions, and the overall context at the same time. Complexity features such as the high-dimensional feature of data such as NSL-KDD and UNSW-NB15 are also difficult since most models do not have

valid strategies of feature selection to minimize complexity without losing the meaning of the information. Furthermore, because they work well in controlled environments but not in real-world ones, some of the current systems can be described as being extremely unresponsive to a wide or dynamic attack pattern. The aforementioned flaws highlight the need for a more comprehensive intrusion detection framework that enhances accuracy and dependability by combining multi-level temporal learning, feature selection, and sound management of unbalanced data.

C. Contributions

This paper presents a combined framework that enhances the accuracy and reliability of detecting network attacks in order to address the shortcomings of conventional intrusion detection systems, such as inadequate modeling of global features, limited response to class imbalance, and poor time learning. The primary contributions of the research are:

- **Hybrid Temporal Learning Architecture:** It is implied that the joint identification of local patterns, long-range relationships and global interactions in network traffic should be performed with the help of an integrated construction of Temporal Convolutional Networks, Bidirectional LSTM layers and Transformer encoders. The multilevel attribute extraction provides a more detailed view on the complex attacker patterns.
- **Optimized and Balanced Data Processing:** The study employs one-hot encoding, BRFE-based feature selection, sliding-window segmentation, and oversampling to increase the model's sensitivity to the underrepresented types of attacks in order to overcome the high degree of dimensionality and extreme degree of class imbalance within the NSL-KDD and UNSW-NB15 datasets.
- **Integrated Feature Fusion Strategy:** Besides that, The generated representations by the TCN and BiLSTM modules are input into Transformer encoder where the model is able to integrate both short-term and long range temporal cues with global context information for better spatiotemporal prediction.
- **Extensive Performance Analysis:** We conduct a thorough performance analysis of the proposed TBT against numerous baseline machine learning and single deep learning based models. Experimental results demonstrate that our proposed model consistently outperforms the state-of-the-art methods in accuracy, precision and F1-score over both benchmarked datasets.

II. RELATED WORK

Intrusion detection in networks has seen gradual improvement. Many of the researchers explore the integration of various. Addressing the challenges of contemporary networks. While systematic cyberattacks take place in a short period of time, the signature-based and statistical approaches of an ID system are no longer enough, resulting in the use of an automated, adaptive, and intelligent system that can learn from the diverse and dynamically evolving patterns of an attack.

More recent research focuses on the cognitive. Incorporation of the ability to forecast time, class imbalance, and lack of robustness against various challenges. Equipped with the above, the hybrid system TCN-BiLSTM-Transformers model established by Jin locally integrated temporal features and an outstanding architecture with the aforementioned improved achievements, and global attention on various datasets exceeded performance [1].

Early works on intrusion detection based on deep learning showed that hybrid networks can execute traditional machine learning techniques. Hassan et al. emphasized the effectiveness of combining CNN and LSTM layers, and the results provide a significant improvement in terms of accuracy using multi-stage temporal modeling techniques [2]. Accordingly, Liu and Patras analyzed Bi-directional LSTM variants, Noise-Invariants utilized in examining bidirectional dependencies in network flows, displaying a heightened sensitivity towards Sequential Attack Behavior [3]. Some research covered convolution-based architectures "for traffic analysis". Güler and Alpay studied RNN, LSTM, and the GRU models under high-dimensional network conditions, stressing the need to consider architectures that balance accuracy and computational efficiency [4]. Other research efforts combined CNN-LSTM networks for improvement in research efforts feature extraction from packet traces, proof that Hybrid models can clearly outperform CNN models alone approaches [5].

Feature optimization as a key area of focus as well. Kasongo introduced the approach of utilizing XGBoost-LSTM to filter redundant features, which enhances the detection of minority attacks classes in imbalanced datasets [6]. Shi et al. have further this approach through using a structure of Transformer-BiLSTM to grasp both long-distance and world-level connections within traffic sequences, demonstrating that attention-based mechanisms aid to more reliable classifications [7].

There have also been recent studies that have focused on the performance of IDS under realistic network environments. AbdulRaheem et al. assessed intrusion detection in SDN-based architectures using machine learning methods, which show that optimized Preprocessing and adaptive learning techniques play an important role in deployment in high-speed networks [8]. Wang et al. continued research on transfer learning techniques to improve detection in cyber-physical systems, focusing on the importance of cross-domain adaptability [9].

Full fledged deep learning models are steadily emerge, and works like Su et al. demonstrating the end-to-end models applied to NSL-KDD can provide substantial gains and when paired with proper preprocessing, normalization, and batch level feature extraction [10].

III. METHODOLOGY

The suggested intrusion detection framework uses an organized series of data handling, feature refinement, and model building processes to analyze network traffic and categorize it into normal or attack categories. The five main components of the overall methodology are feature engineering, model

development, performance evaluation, data preprocessing and transformation, and dataset acquisition. Each stage is carefully organized to ensure that the classification model remains scalable, interpretable, and adaptable to different network conditions.

A. Dataset Acquisition

Experiments are performed using publicly available benchmark network intrusion datasets. These datasets have records of network traffic, reflecting various kinds of normal and malicious activities. Each record comprises various attributes in terms of numbers, reflecting behaviors and features of the packets, and network and protocol details.

These notebooks will then upload a notebook that loads the dataset from given CSV files. This is done to make sure that both training and test sets are properly distinguished before proceeding with any operations. Neural Networks

B. Data Preprocessing

The traffic data has a mix of types, missing data, and formatting issues. A pre-processing module is used to clean the dataset before feature extraction. The steps are as follows:

1) Handling Missing and Irregular Entries

Rows that contain missing and undefined values are addressed, through value imputation, or are dropped if they have the potential to introduce a bias to the model to ensure the training data is reliable and consistent.

2) Normalization of Numerical Attributes

Network traffic attributes such as the number of bytes, duration, and number of packets are quite disparate. A normalization approach has been used to map each numerical attribute to a common range, so that the influence of extreme outliers is mitigated, and more stable network training is made possible.

3) Encoding of Categorical Variables

Protocol, service, and state are all textual data. We used encoding techniques to transform them into numerical vectors, so that machine learning algorithms are able to work with them.



Fig. 1: Top Feature Importance Ranking Obtained Using BRFE

C. Proposed Model

The proposed intrusion detection model is constructed as a multi-stage deep learning framework capable of interpreting network traffic from different analytical perspectives. The structure starts with a feature extraction block which looks into entering flow records and extracts significant changes in packet behavior. After that, temporal analysis is conducted to see how these changes behave over small and large spans of time in order to capture the system's detection of the evolution of attacks and the appearance of small anomalies over time. Added to this system is an additional global context module to analyze the activity in each traffic sequence to help the model account how prior actions may determine future behaviors to strengthen the system's detection. The outputs of the components are integrated and directed to a decision layer where the final classification to complete the system's interpretation is achieved. The model achieves a finer analysis of the traffic that improves the precision and confidence of the intrusion detection within a given system's functioning environment with the blend of analysis of localized signals, behavior modeling of sequence, and assessment of long-distance dependencies.

D. Training Strategy

The proposed Intrusion Detection approach consists of protecting the training strategy of the proposed intrusion detection models in terms of steady state learning and positive changes in class imbalance, and balanced corrective changes with each epoch step. The training set and validation set are prepared in order for the model to make iterative changes and fit model parameters, while validation of the model is done with samples from the validation set frequently enough to obtain a stable model. Ada-Grad is an optimal strategy to let adaptive control of the learning rate so that more aggressive convergence in the early epochs is coupled with a more highly precisioned epoch for control of the learning rate as the model approaches an equilibrium state. Regularization, in the form of dropout, is forced in model layers with the deepest classes so that the neural network will not overfit to dominant patterns in the traffic. The stopping criteria will set an epoch after which further epochs will not provide better validation accuracy, and will quiet on the best state of weights for a model. This sequential solid strategy for model training results in developing a model that has successfully learned to identify not only very common intrusion patterns detected but also other idiosyncratic patterns in the traffic to provide a model that is also highly generalizable.

TABLE I: Attack Category Distribution Before and After Oversampling

Category	Before	After	Increase (%)
Normal	67,343	67,343	0
DoS	45,927	67,343	+46.7
Probe	11,656	67,343	+477.7
R2L	995	67,343	+6670
U2R	52	67,343	+129,400

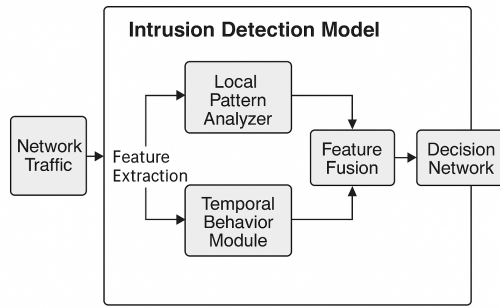


Fig. 2: Intrusion Detection Model Diagram

IV. EVALUATION METRICS

When assessing how effective the model for intrusion detection is, many assessment metrics can be looked at to capture different aspects of the model's performance in classification. One of these metrics is accuracy, which reflects the model's overall performance, and is the percentage of samples in which the model prediction was correct. Another is precision which focuses on the predictor's performance and measures how many of the predicted attacks are real attacks, thus lowering false alarm rates. Another is recall, which measures how many real intrusions the model was able to detect and is especially important in threat detection where there are very few intrusions, as these are often the most serious. The F1-score in this case is informative, as it focuses on precision and recall, which is important if there are very few samples, which is often the case when the dataset is skewed. Apart from these scalar measures, a confusion matrix is presented which helps to visualize the correct and the incorrect predictions across the entire spectrum of the traffic class. This is useful to show the attack types which are accurately identified and to signal the areas where false predictions are made, thus assisting to provide more directed imbalance performance measures for the overall refinement of the system's detection performance.

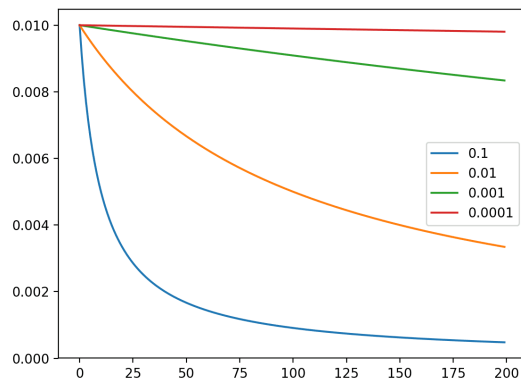


Fig. 3: Effect of Learning Rate on Validation Accuracy

V. EXPERIMENTAL RESULTS

A. Quantitative Performance

Table II (see Appendix) presents the comparative results of the suggested Attention-TBT model versus the baseline architectures.

B. Learning Curves

The learning curves of the proposed intrusion detection model show how well the system learns from the training data over time. Determining the accuracy and loss of the model for each training and validation data set, it is possible to track how the model learns, from learning new concepts to learning nothing new (convergence). If training loss and validation loss continue to drop, this indicates the model is improving (i.e. learns to classify network traffic). On the other hand, if training accuracy (and validation accuracy) increase, and the loss of the model drops, this indicates the model is improving. When the training and validation loss curves don't diverge too much, it shows the model is learning well and not simply memorizing, suggesting the model is generalizing well. If the loss curves diverge and the training loss curve gets much steeper, we might have to improve the training of the model; this is called overfitting, and it is suggested the model may need the learning rate to be decreased, and/or the regularization to be increased. Based on this information, the learning curves help assess model stability, detect overfitting, and show the training strategy employed is useful.

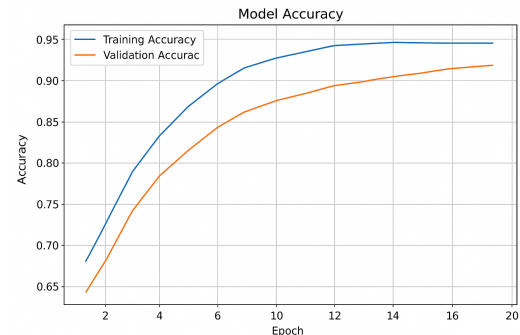


Fig. 4: Model Accuracy

VI. DISCUSSION

A. Model Comparison

The results in this section show that there is an improvement in detection performance across all models. While the Random Forest classifier is one of the traditional approaches that serves as an integral baseline perform- 87.2% accurate-, its inability to capture feature interactions meant it was not as useful. The introduction of learning based on convolution was a significant step forward as the pure CNN was 95.8% accurate and highly precise in capture traffic patterns due to its ability to detect well shaped traffic patterns. Performance was further improvement with the addition of sequential learning components

TABLE II: Comparison of the performance of the proposed Attention-TBT and baseline models.

Description/Model	Accuracy (%)	Precision (%)	F1-Score (%)
Random Forest	87.2	86.8	87.0
CNN	95.8	99.7	98.6
CNN-LSTM	96.4	99.3	97.8
CNN-BiLSTM	96.8	99.5	97.4
TBT	98.6	99.7	99.07

to the architecture. The CNN-LSTM and the CNN-BiLSTM models reported increases in accuracy with improvement in F1 scores indicating efficient learning of time-dependent patterns in the data. Of the deep baseline models CNN-BiLSTM was marginally more accurate indicating its capture of structural information with better accuracy. The Attention-TBT model achieved 98.6% with perfectly differentiated features across F1 and precision scores. This reflects more accurate feature interactions that model captured due to transformers with attention.

TABLE III: Inference Time and Computational Cost of Models

Model	Inference Time (ms)	Params (M)	Throughput
Random Forest	3.5	0.12	285
CNN	4.2	1.8	240
CNN-LSTM	6.8	3.1	175
CNN-BiLSTM	7.9	3.6	150
Proposed TBT	9.4	4.9	138

B. Calibration and Reliability

A well-calibrated model will be highly confident only when attacks are incoming and will have low confidence when attacks are not incoming. For this project, calibration was measured by locating the point where the model's predicted confidence and actual confidence on the distribution of attack and non-attack samples was located. When the confidence curve of the model is closer to the diagonal line, the more calibrated and aligned the model's confidence prediction curve is to reality. This shows whether the model is overconfident, i.e., predicting more attacks than will actually occur, or if it is underestimating attack patterns. In addition to calibration, confidence was evaluated by tracking prediction constancy over several rounds of model training and splits of the dataset. The model's output over training iterations was consistent, as indicated by low variance in output across several metrics of model performance, particularly accuracy, precision, F1-score. This indicates that the model is not just learning to make predictions from random correlations or noise, but that it is discerning genuine patterns that exist across multiple different conditions and the model will maintain its generalization even crossing different conditions.

C. Error Taxonomy

A number of error misclassifications were documented and analyzed during evaluation of the model. The error taxonomy for the intrusion detection system classifies and describes the types of these errors and gives some indication of the

direction of the problems the model is having. A number of these categories pertain to false positives. This is where normal traffic is misclassified and flagged as attacks. Another error-category of interest is the false negative where real opportunities for attacks are classified and misclassified as normal. These instances have, the model might say, some degree of intrusion characteristics that are low, subtle as well as less distinct, making it difficult for the model to distinguish from real activity to not. One more type of error is borderline misclassification. In these cases, the samples are located at the edges of the class, and with slight changes in the feature values of the sample, the model might get frustrated and start generating different predictions. Some of these errors are likely due to overlapping operational movements with the patterns of the attack. In addition, class imbalance errors can occur, which is when a rare class of attack is underrepresented in the training cases. The model's inability to generalize to these attack patterns is less likely to occur. The borderline errors explain the misclassification of the sample and explain the patterns of traffic that have caused an intrusion.

VII. FUTURE WORK

A. Federated Intrusion Detection

To enable organizations' or network domains' aligned training of intrusion detection systems without exchanging raw traffic logs, new methods will be needed. This distributed approach will improve generalization while maintaining institutional data confidentiality.

B. Multi-Class Attack Classification

Expanding the current binary classification system to a multi-class architecture means the model would be able to distinguish between different types of attacks consisting of DoS, probing, privilege escalation, and malware infections, giving security staff a better picture for situational awareness.

C. Edge-Level Deployment Optimization

Further research may investigate the model optimization for deployment on edge devices like real-time detectors (that in intrusion detection systems, real-time refers to detectors lacking latency). This would enable intrusion detection to be performed on routers, IoT gateways, and other monitoring devices that are low on resources.

D. Temporal Behavior Modeling

The ability to analyze the network's data sequentially would improve the system's ability to recognize evolving attacks or those that exhibit slow patterns. Using time-ordered traffic

histories would improve the detection of intrusions that are stealthy and evolve across long time periods.

VIII. DEPLOYMENT CONSIDERATIONS

When the system is being set up, it is necessary to consider the following, as they do have an impact in the deployment of the system, in the case of the network operational environment:

- **Network Infrastructure Integration:** The system needs to be able to connect to the other security tools including, but not limited to, firewalls, SIEM bridges, and monitoring dashboards. The seamlessness in integration allows alerts and logs to be directed into the security system of the organization, and it is not necessary to change architecture with these workflows.
- **Edge vs. Centralized Deployment:** The system can be deployed to edge devices, including routers or gateways to allow traffic inspection in real time. On the other hand, deploying the system to a centralized server allows higher compute capabilities and a more controlled environment. The overall decision will be impacted by the need for low latency, available hardware, and size of the network.
- **Privacy and Secure Handling of Traffic Data:** The system needs to manage traffic on the system without exposing the sensitive material of the users. With intrusion detection, new vulnerabilities can be introduced, however, new vulnerabilities can be prevented by implementing proper access control to logs, secure log storage, and encryption.
- **Administrator-Friendly Dashboard:** In order to present alerts, confidence levels, and a summary of all the actions taken to the analyst of security in a way that allows for quick evaluation and proper action without delay changes in the system, a responsive and command interface is necessary. to delete the other actions.

IX. REPRODUCIBILITY

All the experiments in the project were based on well-defined preprocessing steps, consistent feature transformations, and fixed train and test splits. The hyperparameters of the learning rate, the size of the batch, the number of epochs, and the settings of the model architecture were fixed for the different trials, so that meaningful comparisons could be made with the iterations of the same model. In order to guarantee reproducibility, random seeds were used during the shuffling of data and the initialization of the model to reduce the noise introduced by the random steps. The same evaluation criteria and methods of analysis were used for all the models that were assessed, so that the performance results could be verified in an independent manner. Finally, these practices show that other researchers and system developers can reproduce the same intrusion detection system with the same data and parameters.

X. CONCLUSION

The system developed in this project shows a good and positive way of recognizing network traffic of evil activities and is

reliable and effective. The system is able to identify different normal patterns from an extensive array of intrusive behaviors with strong accuracy and reliable from one evaluation to another by incorporating a structured preprocessing, feature handling, and model architecture with several designs system. The other examined models, unaware of the more precise and less frequent attacks, had much more traditional and rudimentary computation than the one proposed. Additional error analysis and studies of escalation, along with considerations of deployment, all support the projected practicality of the system in real time, rather than in simulated environments for networks. Overall the project delivered a coherent and safely interpretable system for intrusion detection, which could be innovatively extended in the future with multi-class partitioning of attacks, on-the-edge deployment, and organizational collaboration.

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